

STOCHASTIC METHODS IN WATER RESOURCES

Unit 5: Geostatistics

Lecture 18: Other kriging and geostatistical simulation

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Estimating the local conditional distribution

Generalities

- ▶ **Kriging** can also be used to estimate for each **non-observed location** the **probability distribution** $f_z(z; \mathbf{x}|z(\mathbf{x}_i), i = 1, \dots, n)$, i.e the **probability distribution** given the **observed values** at the **observation locations**.
- ▶ Recall that each **realisation** is **conditioned** by the **observations**, i.e. it passes through the observed value, but is free to vary between observations. The farther away from an observation, the more the realisations differ.
- ▶ This is reflected by the **conditional pdf** $f_z(z; \mathbf{x}|z(\mathbf{x}_i), i = 1, \dots, n)$ at a given location. The farther away from an observation, the larger the **variance** of the **conditional pdf**, which means the more **uncertain** we are about the actual but unknown value $z(\mathbf{x})$.
- ▶ The following methods can be used to estimate the **conditional pdf** $f_z(z; \mathbf{x}|z(\mathbf{x}_i), i = 1, \dots, n)$ through kriging.

Estimating the local conditional distribution

Multivariate Gaussian random functions

If, apart from being **wide sense stationary**, the **random space function** is also **multivariate Gaussian distributed** then we have:

1. The **kriging error** is **Gaussian distributed** with **mean** zero and **variance** equal to the **simple kriging variance** $\sigma_{SK}^2(\mathbf{x}_0) = \text{var} [\hat{Z}(\mathbf{x}_0) - Z(\mathbf{x})]$. A **95%-prediction interval** would then be given by $[\hat{z}_{SK}(\mathbf{x}_0) - 2\sigma_{SK}(\mathbf{x}_0), \hat{z}_{SK}(\mathbf{x}_0) + 2\sigma_{SK}(\mathbf{x}_0)]$ where $\hat{z}_{SK}(\mathbf{x}_0)$ is the **simple kriging prediction**.
2. The **conditional cumulative probability distribution function (ccpdf)** is **Gaussian** with **mean** equal to the **simple kriging prediction** $\hat{z}_{SK}(\mathbf{x}_0)$ and **variance** equal to the **variance of the simple kriging prediction error** $\sigma_{SK}^2(\mathbf{x}_0)$:

$$F_{z|z_1, \dots, z_n}(z; \mathbf{x}_0 | z(\mathbf{x}_i), i = 1, \dots, n) = \frac{1}{\sqrt{2\pi\sigma_{SK}^2(\mathbf{x}_0)}} \int_{-\infty}^z e^{-\frac{[z - \hat{y}_{SK}(\mathbf{x}_0)]^2}{\sigma_{SK}^2(\mathbf{x}_0)}} dz \quad (1)$$

where $z(\mathbf{x}_1), \dots, z(\mathbf{x}_n)$ are the observed values at locations $\mathbf{x}_1, \dots, \mathbf{x}_n$ respectively.

The second property is very convenient and the reason why the **multivariate Gaussian** and **stationary random spatial function** is very popular model in **geostatistics**.

Estimating the local conditional distribution

Multivariate Gaussian random functions

- ▶ After performing **simple kriging predictions**, one is able to give an estimate of the **ccpdf** of $Z(\mathbf{x})$ **exceeding** a given **threshold** for every location in the domain of interest.
- ▶ For instance, if $Z(\mathbf{x})$ is a **concentration of some pollutant** in the groundwater and z_c is **critical threshold** above which the **pollutant** becomes a **health hazard**, **simple kriging** and equation 1 can be used to map the **probability of exceeding** this **threshold**, given the concentrations found at the observation locations.
- ▶ Instead of **delineating** a **single plume** based upon some **predicted value**, **several alternative plumes** can be delineated, depending on which **probability contour** is taken as its boundary.
- ▶ This way, both the **observed concentration** values as well as the **local uncertainty** are taken into account when mapping the plume.
- ▶ Also, the **risk** of not treating **hazardous groundwater** can be weighted against the **costs of remediation**. For instance, if the risk of not treating **hazardous groundwater** should be smaller than 5%, all the water within the 0.05 contour should be treated.
- ▶ Obviously this results in much higher costs then if, for instance, a 10% **risk** is deemed acceptable.

Estimating the local conditional distribution

Log-normal kriging

- ▶ Many geophysical variables, such as **hydraulic conductivity** and **pollutant concentration**, are approximately **log-normal distributed**.
- ▶ A frequently used **random space function** model to describe these variables is the **multivariate log-normal distribution**.
- ▶ If $Z(\mathbf{x})$ is **multivariate log-normal distributed**, the natural logarithm $Y(\mathbf{x}) = \ln(Z(\mathbf{x}))$ is **multivariate normal distributed**. **Log-normal kriging** then consists of the following steps:
 1. Transform the observations $z(\mathbf{x}_i)$ by taking their logarithms $y(\mathbf{x}_i) = \ln(z(\mathbf{x}_i))$.
 2. Estimate the **semivariogram** $\hat{\gamma}_Y(\mathbf{h})$ from the log-transformed data $y(\mathbf{x}_i)$ and fit a permissible model (note: that mean m_Y must be determined and assumed known if **simple kriging** is used).
 3. Using the **semivariogram** $\hat{\gamma}_Y(\mathbf{h})$, the data $y(\mathbf{x}_i)$ (and the mean m_Y in case of **simple kriging**), the kriging equations are solved to obtain at every non-observed location $\mathbf{x}_0$ the **prediction** $\hat{Y}_{SK}(\mathbf{x}_0)$ and **prediction error variance** $\sigma_{Y_{SK}}^2(\mathbf{x}_0)$ in case of **simple kriging** or $\hat{Y}_{OK}(\mathbf{x}_0)$, $\sigma_{Y_{OK}}^2(\mathbf{x}_0)$ in case of **ordinary kriging**.
 4. An unbiased prediction of $Z(\mathbf{x}_0)$ is obtained by the following backtransforms footnote:
 - ▶ for **simple kriging**:

$$\hat{Z}(\mathbf{x}_0) = e^{\hat{Y}_{SK}(\mathbf{x}_0) + \frac{1}{2} \sigma_{Y_{SK}}^2} \quad (2)$$

- ▶ for **ordinary kriging**:

$$\hat{Z}(\mathbf{x}_0) = e^{\hat{Y}_{OK}(\mathbf{x}_0) + \frac{1}{2} \sigma_{Y_{OK}}^2 - \nu_Y} \quad (3)$$

where ν_Y is the **lagrange multiplier** used in the **ordinary kriging** system.

Estimating the local conditional distribution

Log-normal kriging

5. If $Y(\mathbf{x})$ is **multivariate Gaussian** distributed and **stationary**, the **conditional cumulative probability distribution function (ccpdf)** can be calculated from the **simple kriging** prediction $\hat{y}_{SK}(\mathbf{x}_0)$ and prediction **variance** as:

$$F_{z|z_1, \dots, z_n}(z; \mathbf{x}_0 | z(\mathbf{x}_i), i = 1, \dots, n) = \frac{1}{\sqrt{2\pi\sigma_{Y_{SK}}^2(\mathbf{x}_0)}} \int_{-\infty}^z e^{-\frac{[\ln(z) - \hat{y}_{SK}(\mathbf{x}_0)]^2}{\sigma_{Y_{SK}}^2(\mathbf{x}_0)}} dz \quad (4)$$

An additional reason why in many geostatistical studies the observations are **log-transformed** before kriging is that the **semivariogram** of log-transforms can be better estimated (shows less noise) because of the imposed **variance** reduction.

Estimating the local conditional distribution

Kriging normal-score transforms

An even more general **transformation** is the **normal-score transform** using the **histogram**. Through this transform, it is possible to **transform** any set of observations to **univariate Gaussian distributed variables**, regardless of the distribution of these observations. A **normal-score transform** proceeds as follows:

1. The n observations are **ranked** in ascending order:

$$z(\mathbf{x}_i)^{(1)} \leq z(\mathbf{x}_j)^{(2)} \leq \dots \leq z(\mathbf{x}_k)^{(r)} \leq \dots \leq z(\mathbf{x}_l)^{(n)}$$

where $r = 1, \dots, n$ are the **rank**s of the observations.

2. The **cumulative probability** associated with observation $z(\mathbf{x}_k) = z_k$ with rank r is estimated as:

$$\hat{F}(z_k) = \frac{r(z_k)}{n + 1} \quad (5)$$

or in case of **declustering**

$$\hat{F}(z_k) = \sum_{i=1}^{r(z_k)} w_{r(z_k)} \quad (6)$$

3. The associated **normal score transform** is given by the $\hat{p}_r$ -quantile of the **standard normal distribution**:

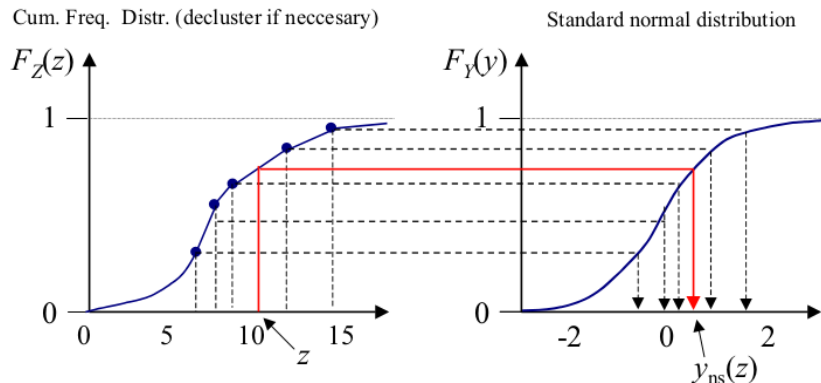
$$y_{ns}(z_k(\mathbf{x}_k)) = N^{-1} \left[\hat{F}(z_k(\mathbf{x}_k)) \right] \quad (7)$$

where $N(z)$ is the **standard Gaussian cumulative distribution function** and $N^{-1}(p)$ its **inverse**.

Estimating the local conditional distribution

Kriging normal-score transforms

The figure below shows graphically how the **normal-score transform** works. The left figure shows the estimated **cumulative distribution** of the original (non-transformed) data and the right figure the **standard Gaussian cumulative distribution**. The dashed lines show how the observed values z_k are transformed into the **normal-score transform** $y_{ns}(z_k)$.



Estimating the local conditional distribution

Kriging normal-score transforms

If we assume that the normal-score transforms are stationary and multivariate Gaussian distributed, the local ccpdfs can be obtained through simple kriging as follows:

1. Perform a normal-score transform of the observations such as described above.
2. Estimate the semivariogram of the normal-score transformed data $y_{ns}(\mathbf{x}_k) = y_{ns}(z(\mathbf{x}_k))$ and fit a permissible semivariogram model $\gamma_Y(\mathbf{h})$. By definition, the mean value of the transformed random space function $Y_{ns}(\mathbf{x}_0)$ is zero.
3. Use the fitted semivariogram model and the normal-score transforms $y_{ns}(\mathbf{x}_k)$ in the simple kriging equations (with $m_Y = 0$) to obtain the prediction $\hat{Y}_{SK}(\mathbf{x}_0)$ and the associated prediction error variance $\sigma_{Y_{SK}}^2(\mathbf{x}_0)$.
4. The local ccpdf is then given by

$$\begin{aligned} F_{z|z_1, \dots, z_n}(z; \mathbf{x}_0 | z(\mathbf{x}_i), i = 1, \dots, n) \\ = Pr[G(\hat{Y}_{SK}(\mathbf{x}_0), \sigma_{Y_{SK}}(\mathbf{x}_0)) \leq y_{ns}(z)] \\ = \frac{1}{\sqrt{2\pi\sigma_{Y_{SK}}^2(\mathbf{x}_0)}} \int_{-\infty}^z e^{\frac{-[y_{ns}(z) - \hat{Y}_{SK}(\mathbf{x}_0)]^2}{\sigma_{Y_{SK}}^2(\mathbf{x}_0)}} dz \end{aligned} \quad (8)$$

where $y_{ns}(z)$ is the normal-score transform of the value z and $G(\mu, \sigma)$ a Gaussian variate with mean μ and variance σ .

This is also shown graphically in figure above. Suppose we want to know for the non-observed location the probability that $Z(\mathbf{x}_0) \leq z$. We first obtain through the transformation the value of $y_{ns}(z)$. From the simple kriging of transformed data we have at $\mathbf{x}_0$: $\hat{Y}_{SK}(\mathbf{x}_0)$ and $\sigma_{Y_{SK}}^2(\mathbf{x}_0)$. Finally, we evaluate $Pr[G(\hat{Y}_{SK}(\mathbf{x}_0), \sigma_{Y_{SK}}(\mathbf{x}_0)) \leq y_{ns}(z)]$ (see equation 8) to obtain the probability.

Estimating the local conditional distribution

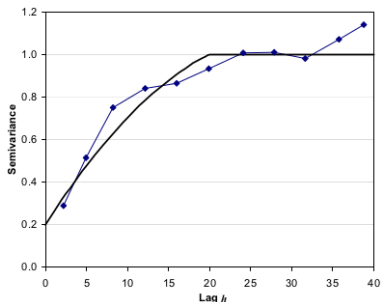
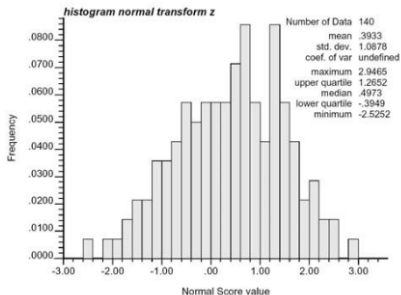
Kriging normal-score transforms

- ▶ To calculate the **normal-score transform** of any given value z (which is not necessarily equal to the value of one of the observations), the resolution of the estimated **cdf** $\hat{F}(z)$ must be increased.
- ▶ Usually, a **linear interpolation** is used to estimate the values of $\hat{F}(z)$ between two observations (see figure above). Of course, most critical is the **extrapolation** that must be performed to obtain the **lower** and **upper tails** of $\hat{F}(z)$.
- ▶ For instance, if the **upper tail** of $\hat{F}(z)$ rises too quickly to 1, the probability of high values of z (e.g. a **pollutant in groundwater**) may be underestimated.
- ▶ Usually a **power model** is used to **extrapolate** the **lower tail** and a **hyperbolic model** to **extrapolate** the **upper tail**.

Estimating the local conditional distribution

Kriging normal-score transforms

The figure below shows the **histogram** and the **semivariogram** of the **normal score transforms**. It can be seen that **semivariogram** is less noisy than that of the **non-transformed data**, because transformation decreases the effect of the very large values.

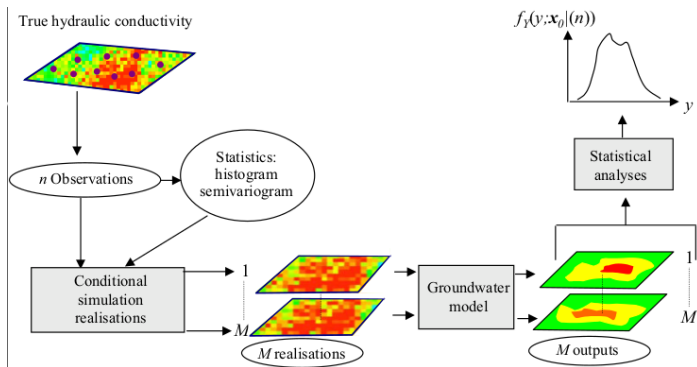


Generalities

- ▶ The third field of application of **geostatistics** is simulating realisations of the **conditional random function** $Z(\mathbf{x}|z(\mathbf{x}_i), i = 1, \dots, n)$.
- ▶ The aim of **geostatistical simulation** is to generate in the individual conditional realisations. There are two important reasons why sometimes individual realisations of the conditional **random spatial function** are preferred over the interpolated map that is provided by **kriging**:
 1. **kriging** provides a so called **best linear prediction** (it produces values that minimize the **variance of the prediction error**: $\text{var}[\hat{Z}(\mathbf{x}_0) - Z(\mathbf{x}_0)]$), but the resulting maps are much smoother than reality. The individual realisations are very noisy and rugged while the **kriging prediction** produces a smoothly varying surface. The **noisy realisations** have a **semivariogram** that resembles that of the data, so one can say that the real variation of the property considered is much more like the realisations than the **kriging map**.
 2. **multiple realisations** as input for a model can be used for **uncertainty analysis** and **ensemble prediction**. We only have limited information about reality and we therefore represent our uncertainty about reality with a random function. Returning to the example of **hydraulic conductivity**, if we are uncertain about the **parameters** of a **groundwater model**, we also want to know what the uncertainty is about the model output (heads, fluxes). So instead of analysing a single input of **hydraulic conductivity**, a large number of conditional realisations of **hydraulic conductivity** (say 1000) are used as model input. The **variance** of the output realisations is a measure of our **uncertainty** about the output (e.g. **hydraulic heads**) that is caused by our **uncertainty** (lack of perfect knowledge) about the **model parameters** (e.g. **hydraulic conductivity**).

Generalities

The figure is a schematic representation of **Monte Carlo simulation** applied for **uncertainty analysis** of **hydraulic conductivity** in **groundwater modelling**. **Hydraulic conductivity** is spatially varying and sampled at a limited number of locations. **Hydraulic conductivity** is modelled as a **random space function**. Using the observations statistics are estimated that characterise this function (**histogram**, **semivariogram**). Next M realisations of this **random function** are simulated and used in the **groundwater model**. This yields M realisations of **groundwater model** output (e.g. **head fields**). From these realisations it is possible to obtain for a given location (e.g. $\mathbf{x}_0$) the **probability density function** of the output variables (e.g. **head**, **concentration**).



Generalities

For the generation of multiple realisations (**Monte Carlo simulations**) of the **conditional random space function**, commonly referred to as (conditional) **geostatistical simulation**. There are quite a few methods for **simulating realisations** of **MultiGaussian random space functions**. The most commonly used are **LU-decomposition**, the **turning band method** and **Sequential Gaussian simulation**, while there are even more methods for **simulating non-Gaussian random functions**. The **most flexible simulation algorithm** and mostly used nowadays is **sequential simulation**. **Sequential Gaussian simulation (sGs)** will be treated here briefly. **Conditional simulation** with **sGs** needs the **mean** μ_Z and the **semivariogram** of the **random space function** $\gamma_Z(h)$ and proceeds as follows.

1. The **area** is divided into a finite number of grid points N (location indices $\mathbf{x}_1, \mathbf{x}_2, \dots, \mathbf{x}_N$) at which values of the conditional realisations are to be simulated. The grid points are visited in a random order.
2. For the first grid point $\mathbf{x}_1$ a **simple kriging** is performed from the given data $z(\mathbf{s}_1), \dots, z(\mathbf{s}_n)$ yielding the prediction $\hat{Z}_{SK}(\mathbf{x}_1)$ and the **prediction variance** $\sigma_{SK}^2(\mathbf{x}_1)$. Under the assumption that $Z(\mathbf{x})$ is **stationary and multiGaussian** the **conditional cumulative distribution** is **Gaussian**:

$$F_Z(z, \mathbf{x}_1 | z(\mathbf{s}_1), \dots, z(\mathbf{s}_n)) = N \left[z; \hat{Z}_{SK}(\mathbf{x}_1), \sigma_{SK}(\mathbf{x}_1) \right] \quad (9)$$

Generalities

3. A random value P between zero and one is drawn from a **uniform distribution** $U[0, 1]$. Using the **inverse of the conditional distribution** (see equation 9) the **random quantile** P is used to draw a **random value** Z :

$$Z(\mathbf{x}_1) = N^{-1} \left[P; \hat{Z}_{SK}(\mathbf{x}_1), \sigma_{SK}(\mathbf{x}_1) \right] \quad (10)$$

4. For the second grid point $\mathbf{x}_2$ a **simple kriging** is performed using the data $z(\mathbf{s}_1, \dots, \mathbf{s}_n)$ and the previously simulated value $z(\mathbf{x}_1)$ in the **kriging equations** (so the previously simulated value is now treated as a data point). This yields the prediction $\hat{Z}_{SK}(\mathbf{x}_2)$ and the prediction $\sigma_{SK}^2(\mathbf{x}_2)$ from which the **conditional cumulative distribution**

$F_Z(z, \mathbf{x}_2 | z(\mathbf{x}_1), z(\mathbf{s}_1), \dots, z(\mathbf{s}_n)) = N \left[z; \hat{Z}_{SK}(\mathbf{x}_2), \sigma_{SK}(\mathbf{x}_2) \right]$ is build.

5. A random value P between zero and one is drawn from a **uniform distribution** $U[0, 1]$ and using the **inverse of the conditional distribution**

$N^{-1} \left[P; \hat{Z}_{SK}(\mathbf{x}_2), \sigma_{SK}(\mathbf{x}_2) \right]$ the **random quantile** P is used to draw a random value $Z(\mathbf{x}_2)$.

6. For the third grid point $\mathbf{x}_3$ a **simple kriging** is performed using the data $z(\mathbf{s}_1, \dots, \mathbf{s}_n)$ and the previously simulated values $z(\mathbf{x}_1)$, $z(\mathbf{x}_2)$ in the **kriging equations** yielding

$F_Z(z, \mathbf{x}_3 | z(\mathbf{x}_1), z(\mathbf{x}_2), z(\mathbf{s}_1), \dots, z(\mathbf{s}_n)) = N \left[z; \hat{Z}_{SK}(\mathbf{x}_3), \sigma_{SK}(\mathbf{x}_3) \right]$.

Generalities

7. Using a random value P drawn from a uniform distribution $U[0, 1]$ the random variable $Z(\mathbf{x}_3)$ is drawn and added to the data set.
8. Steps 6 and 7 are repeated adding more and more simulated values to the conditioning data set until all values on the grid have been simulated: the last simple kriging exercise thus yields the conditional probability $F_Z(z, \mathbf{x}_N | z(\mathbf{x}_1), \dots, z(\mathbf{x}_N), z(\mathbf{s}_1) \dots, z(\mathbf{s}_n))$.

It can be shown heuristically that by construction this procedure produces a draw (realisation) from the multivariate conditional distribution $F_Z(z(\mathbf{x}_1), \dots, z(\mathbf{x}_N) | z(\mathbf{s}_1) \dots, z(\mathbf{s}_n))$, i.e. a realisation from the conditional random function $Z((\mathbf{x}) | z(\mathbf{s}_1) \dots, z(\mathbf{s}_n))$. To simulate another realisation the above procedure is repeated using a different random path over the grid nodes and drawing different random numbers for the quantiles $P \sim N[0, 1]$. Unconditional realisations of the random function $Z(\mathbf{x})$ can also be simulated by starting at the first grid point with a draw from the Gaussian distribution $N\{z; \mu_Z, \sigma_Z\}$ and conditioning at every step on previously simulated points only. Obviously, the number of conditioning points and thus the size of the kriging system to be solved increases as the simulation proceeds. This would lead to unacceptably large computer storage requirements and computation times. To avoid this, a search area is used, usually with a radius equal to the semivariogram range, while only a limited number of observations and previously simulated points in the search radius are used in the kriging system.

Generalities

The assumption underlying the simulation algorithm is that the random spatial function $Z(\mathbf{x})$ is stationary and multiGaussian. For a random spatial function to be multiGaussian it should at least have a univariate Gaussian distribution $f_Z = N\{z; \mu_Z, \sigma_Z\}$. So, if this method is applied, for instance, a normal score transformation is required. The simulation procedure for a realisation of $Z((\mathbf{x})|z(\mathbf{s}_1), \dots, z(\mathbf{s}_n))$ would then involve the following steps:

1. Perform a normal score transform of the observations
$$y_{ns}(z_k(\mathbf{s}_i)) = N^{-1} \left[\hat{F}(z_k(\mathbf{x}_i)) \right].$$
2. Estimate the semivariogram of the normal-score transformed data $y_{ns}(\mathbf{x}_i)$ and fit a permissible semivariogram model $\gamma_Y(\mathbf{h})$. By definition, the mean value of the transformed random spatial function $Y_{ns}(\mathbf{x})$ is zero.
3. Assuming $Y_{ns}(\mathbf{x})$ to be stationary and multiGaussian, simulate a realisation of the conditional random function $Y_{ns}(\mathbf{x}|y_{ns}(\mathbf{x}_1), \dots, y_{ns}(\mathbf{x}_N))$ using sequential Gaussian simulation.
4. Back-transform the simulated values ($z(\mathbf{x}) = \hat{F}^{-1} [N\{y_{ns}(z_k(\mathbf{x}))\}]$) to obtain a realisation of the conditional random function $Z((\mathbf{x})|z(\mathbf{s}_1), \dots, z(\mathbf{s}_n))$.